Faculty of Sciences of the
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Master's Degree in Information Security

Systems & Data Security

Public Ledger for Auctions over Kademlia

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Abstract

In this report, we briefly describe the application we developed and discuss the decisions we made.

Contents

List of Figures

Acronyms

P2P Peer-To-Peer

PoS Proof Of Stake

RPC Remote Procedure Call

1 Introduction

Kademlia [1]

This report is structured as follows: §5 discusses some challenges that come with the potential implementation of a reputation system. Finally, §7 concludes the report.

2 Problem & Proposed Solution

Similar to Peermart [2].

Our implementation is divided in three parts:

– **P2P network:**

– **Public Ledger:** The public ledger should be able to store auction transactions.

– **Auction Service:**

3 Kademlia Network

In addition to the four RPC of the Kademlia protocol, we implemented two additional RPCs: one to broadcast a message to all the nodes in the network, and one to

3.1 Sybil & Eclipse Attacks

[3].

4 Public Ledger

Regarding PoS, we implemented

5 Reputation System

Due to time limitations, we were not able to implement a reputation system. Regardless, our idea was to implement a system that allowed us to identify unreliable nodes within the network. As a consequence, these nodes would be used less often for routing and would potentially be removed (ignorados pq nao da para remover um nodo ?) altogether from the network.

To achieve this, we would compute and publish a `reputation` score to each node in the network, which would be then updated based on the behavious of the corresponding node – deviating from the protocol would lead to a decrease in reputation, whereas adherence to the protocol would result in an increase in reputation. While implementing such a system may seem a simple task, it is by no means a trivial task due to several intricacies.

For example, there is the possibility of a node accumulating excessive positive reputation, which would potentially allow it to perform negative actions without any penalty. To address this issue, different approaches can be considered. We could, for example, implement a system where it becomes increasingly difficult to gain reputation for nodes with high and low reputation, which would make it harder to accumulate reputation and more challenging to recover from negative actions. On the other hand, we could also cap the maximum reputation value to prevent this issue. Finally, we could implement a system where the penalty for malicious actions increases with a higher reputation score. Idealy, a combination of these approaches should be considered.

In addition, such a reputation system should balance accuracy and efficiency. In other words, it should be able to accuratly capture a node's behavior while maintaining low overheads in terms of computation, communication and storage. Finally, it should also be designed to prevent collusion.

6 Auction System

7 Conclusion & Future Work

Unfortunately, due to timing constraints/time limitations, we were not able to test our application using Google Cloud's virtual machines. Nevertheless,

... using an identity-based approach, such as the one discussed in [4].

References

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[2] David Hausheer and Burkhard Stiller. Peermart: The technology for a distributed auction-based market for peer-to-peer services. In *IEEE International Conference on Communications, 2005. ICC 2005. 2005*, volume 3, pages 1583–1587. IEEE, 2005.

[3] Riccardo Pecori. S-Kademlia: A trust and reputation method to mitigate a Sybil attack in Kademlia. *Computer Networks*, 94:205–218, 2016.

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